HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information and communications technology

Software Requirement Specification

Version 1.2

AIMS: An Internet Media Store

Subject: Software Requirement Specification

TEAM-06

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1 Introduction

1.1 Objective

This SRS states what the AIMS software must do so designers can build and testers can verify it. It specifies externally observable behavior for Customers, Product Managers, and Administrators, plus interactions with external services (payments, email).

1.2 Scope

Product name: AIMS — An Internet Media Store.

In scope (high level): browsing/searching products without login; cart management; placing and paying for orders (QR / credit-card); order confirmation/cancellation and email notifications; pending-order processing by Product Managers; product maintenance (add/update/delete) with history and business rules; user/role administration and notification emails.

Out of scope: UI design details (deferred to UI specs/prototypes per slides).

1.3 Glossary

No	Term	Explanation	Example	Note
1	AIMS	The software being specified	—	System under development
2	Customer	Public user (no login) who browses, carts, and orders	—	No login required.
3	Product Manager (PM)	Staff who maintains products and processes pending orders	—	Role-based access.
4	Administrator	Staff who manages users/roles and admin actions	—	Sends admin emails.
5	VAT	Value-Added Tax applied to product totals/invoice	—	Appears on invoice.
6	Pending order	Order awaiting PM approval/rejection	—	30 per page.
7	CCU	Concurrent users (non-functional throughput metric)	—	Targets below.

8	VietQR / PayPal Sandbox	Payment services used for QR / credit-card payment and refund flows	QR code and REST APIs	Demo integrations.
9	API	Applications programming interface - a set of protocols that specifies the way applications communicate with each other	VietQR API	

1.4 References

- AIMS-ProblemStatement-ver3.0.pdf
- 02-RequirementModelingWithUC.pdf

2 Overall Description

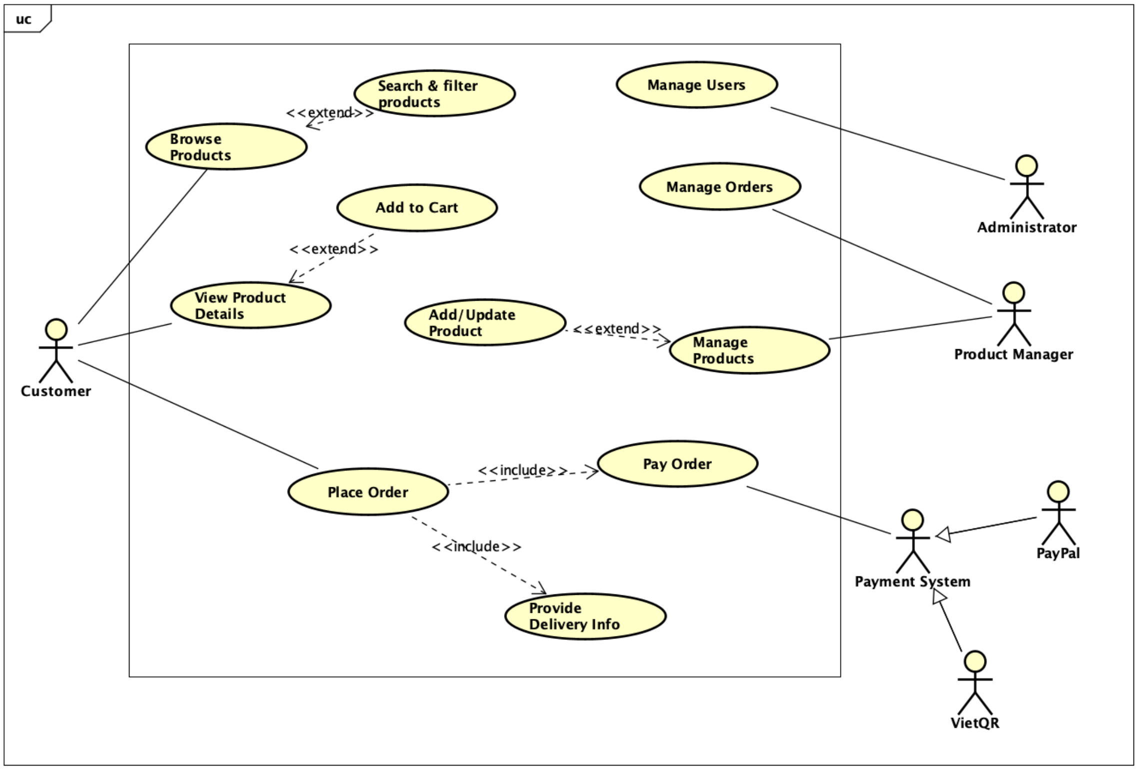
2.1 Survey

List of actors:

- **Customer (no login):** Public user who browses/searches products, manages a cart, and goes through ordering and payment without authentication. Can pay by default via QR and optionally switch to credit card during checkout.
- **Product Manager:** Staff user (role-based access) responsible for catalog maintenance (add/update/delete with recorded history and business rules) and for reviewing pending orders to approve or reject.
- **Administrator:** Staff user (role-based access) who manages users and roles: create/view/update/delete, reset passwords, block/unblock, set roles; the system sends administrative notification emails.
- **VietQR (external payment service):** Third-party QR payment provider used by AIMS for default checkout: AIMS calls VietQR to generate a QR code, displays it, and verifies the transaction via callbacks/sync. Sandbox/test endpoints and callback specs are provided for integration.
- **PayPal (external payment service):** Credit-card payment and refund provider used in AIMS for demo purposes: integrates via PayPal REST APIs (Payments/Orders/Capture/Refund); refunds are issued through the API for captured payments.

Key data: products (with type-specific attributes), carts, orders, users/roles, emails, payments, shipping fees.

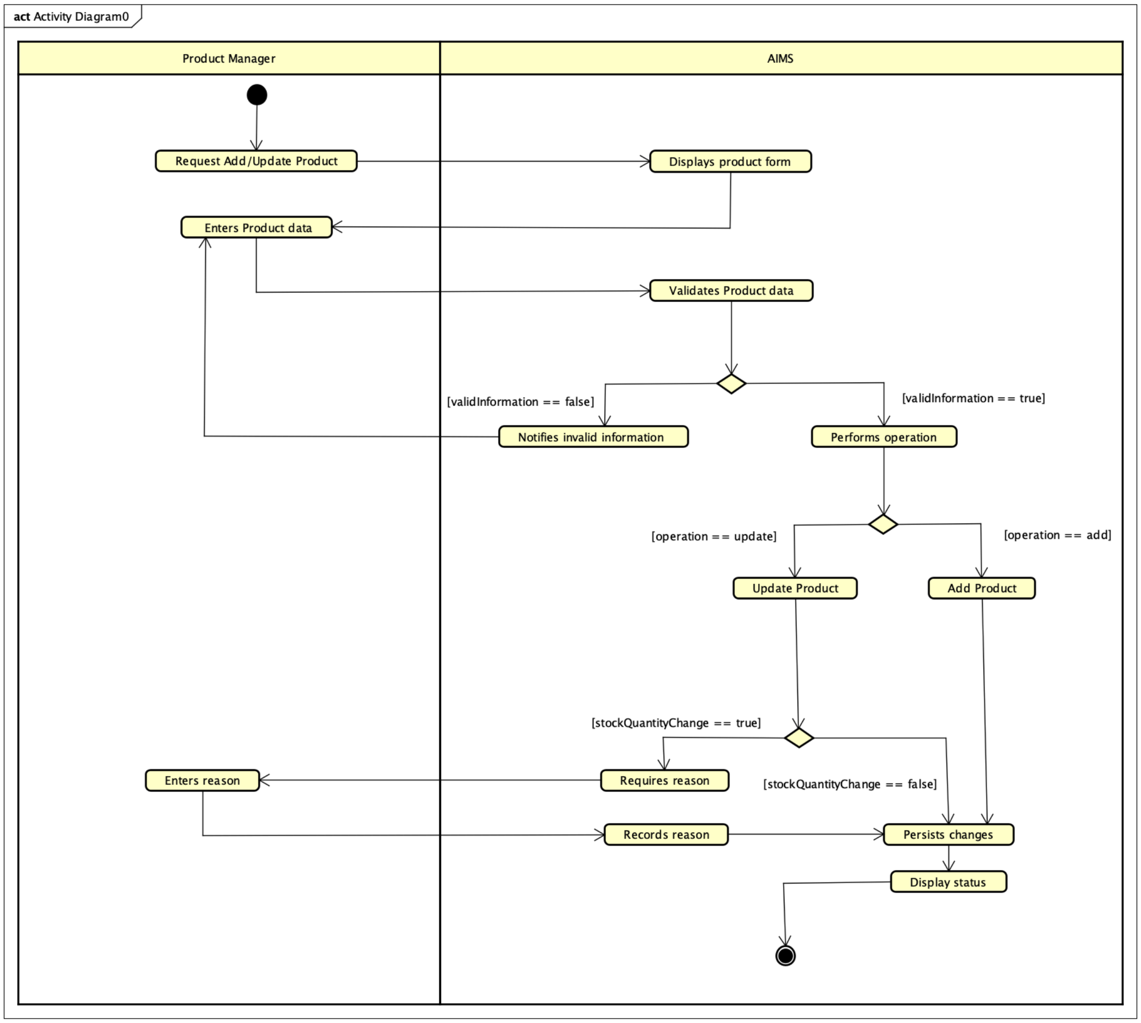
2.2 *Overall requirements*



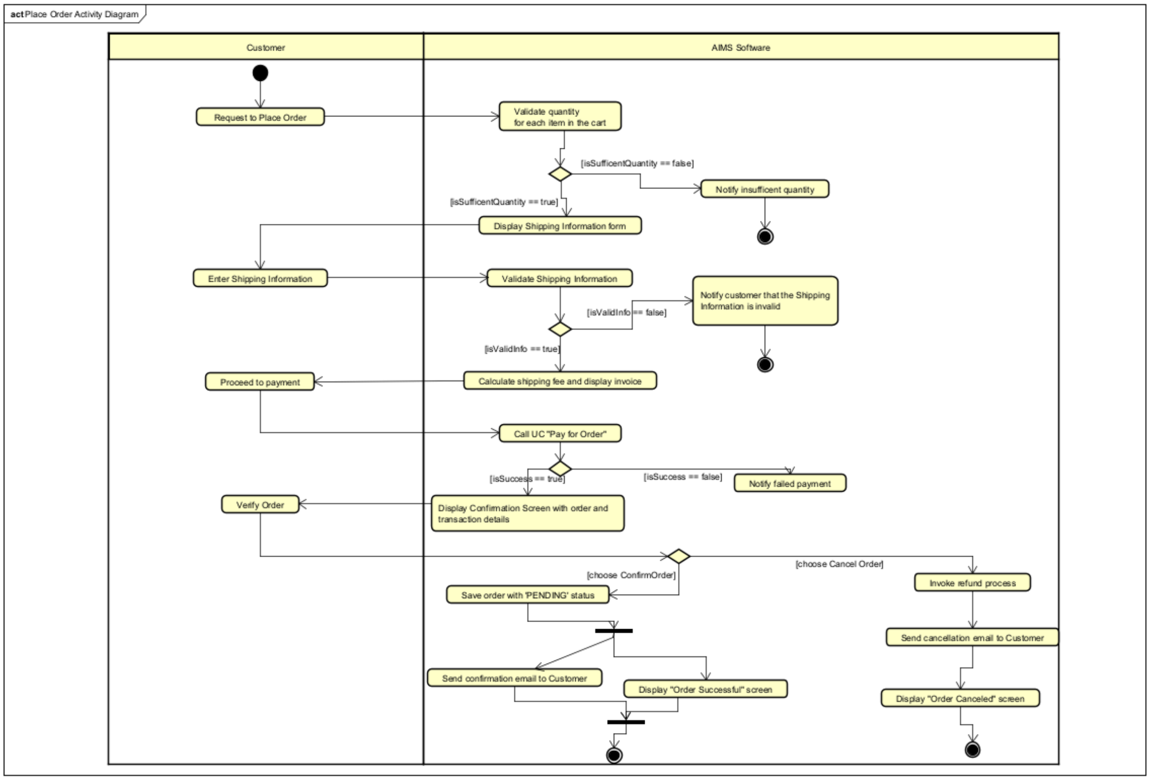
2.3 *Business process*

- Customer can browse a list (20 random products on start), search by title/category, filter by price ranges, and view product details by type.
- Customer can add items to cart, view the cart (totals excl. VAT, itemized details), change quantities/remove items, and be notified of insufficient stock and lacking quantity.
- When placing an order, the system checks stock; collects/validates delivery info; calculates shipping fees (weight/location/rules); shows invoice (incl. VAT & shipping); supports QR or credit-card payment; then confirmation/cancellation with emails and refund; records payment and placed order.
- Product Managers maintain products (add/update/delete) with history and business rules (price in 30–150% of original; notify on invalid input/date). They process pending orders (approve/reject, 30 per page, rejection emails).
- Administrators manage users/roles, block/unblock, reset passwords, and the system sends admin emails; role-based login is required for Admin/PM features.

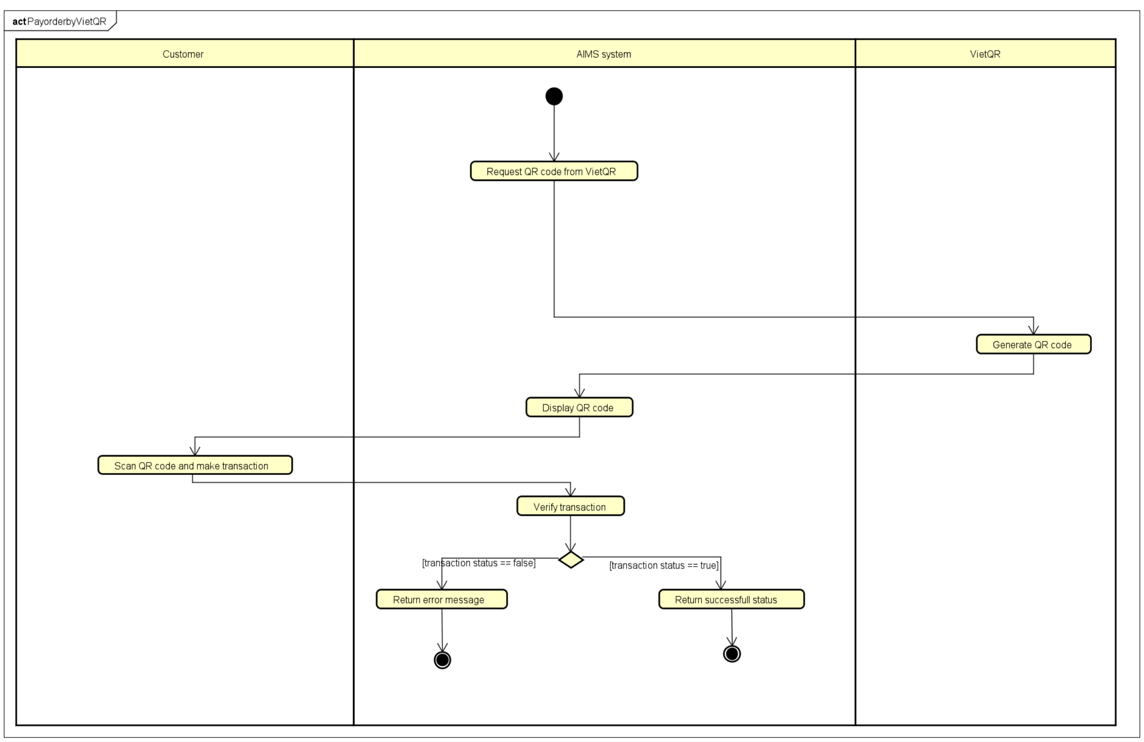
2.3.1. Add/Update Product



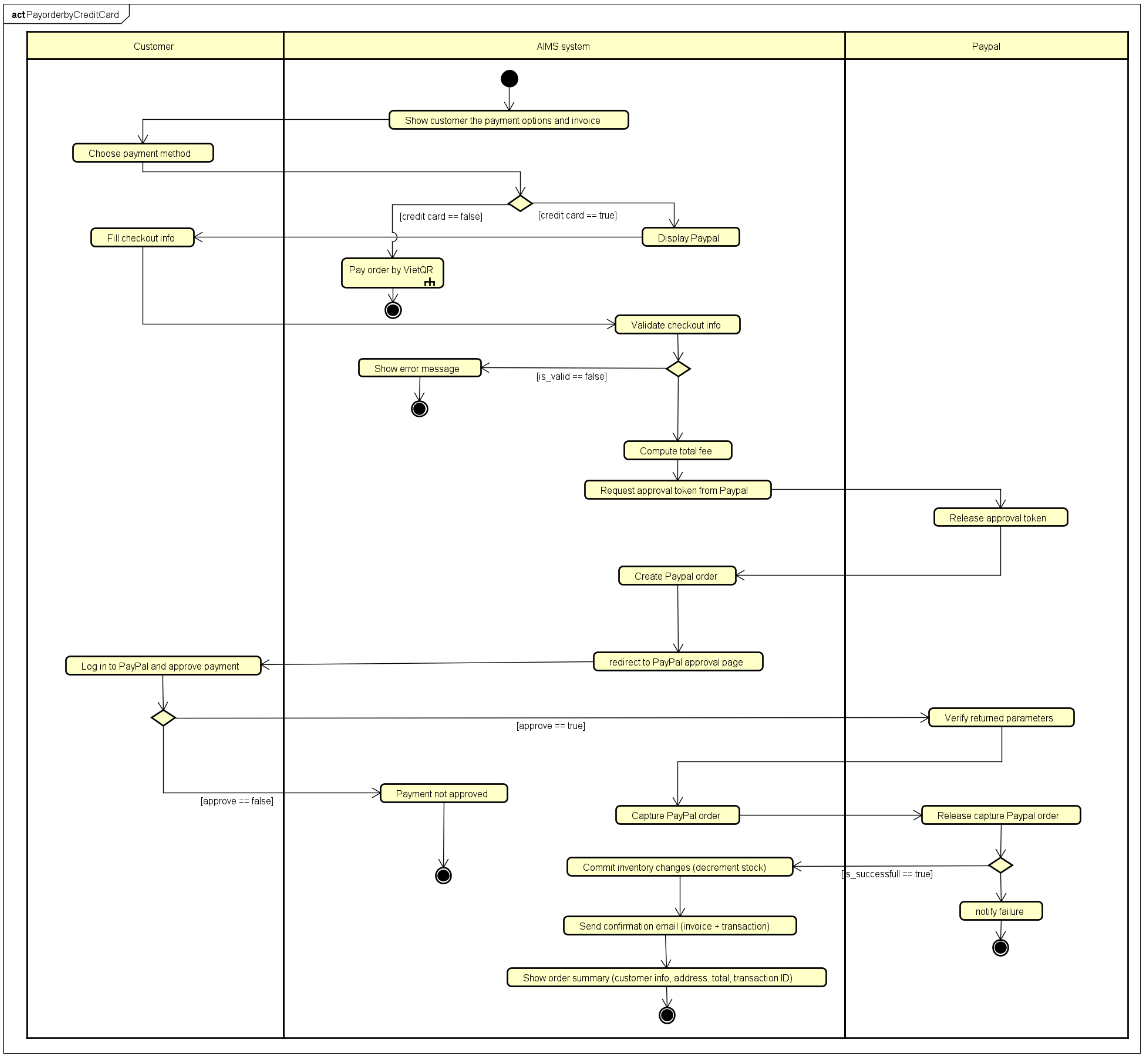
2.3.2. Place Order



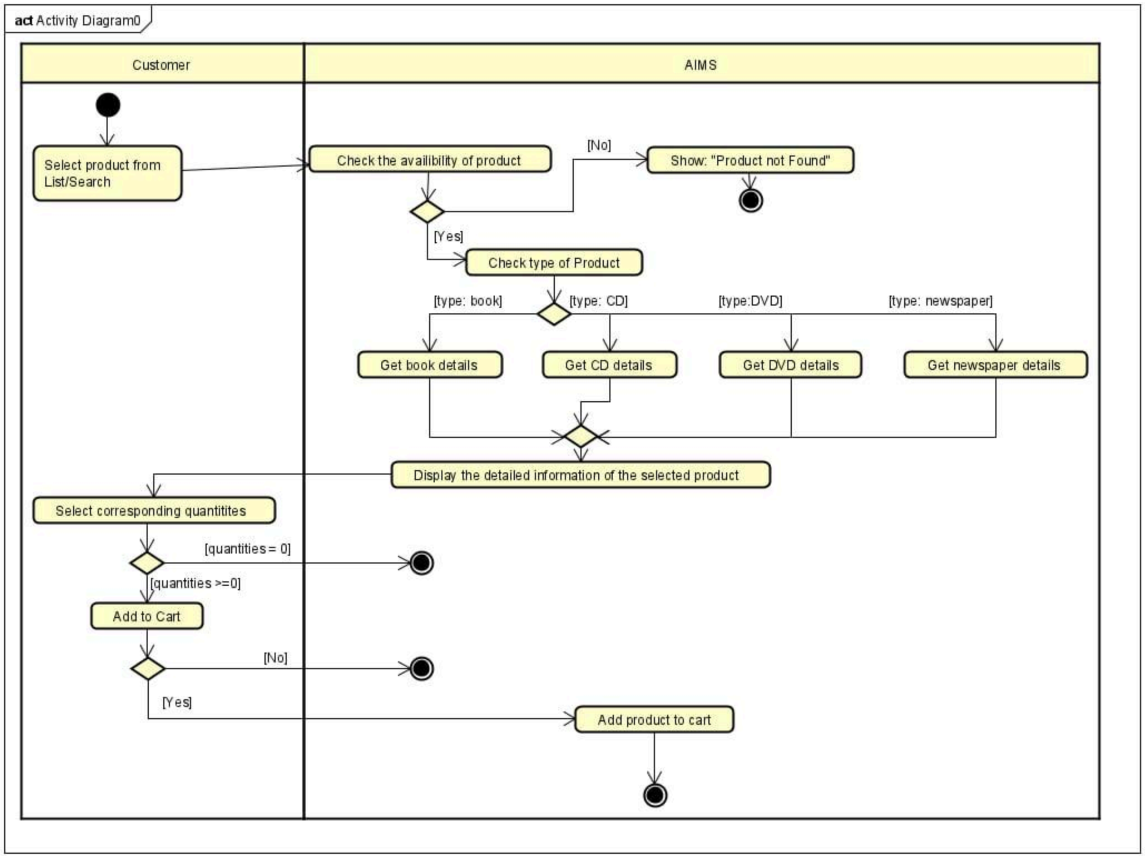
2.3.3. Pay order by VietQR



2.3.4. Pay order by Credit Card



2.3.5. View Product Details


act Activity Diagram0
Customer
AIMS
Select product from List/Search
Check the availibility of product
[No]
Show: "Product not Found"
[Yes]
Check type of Product
[type: book]
[type: CD]
[type:DVD]
[type: newspaper]
Get book details
Get CD details
Get DVD details
Get newspaper details
Display the detailed information of the selected product
Select corresponding quantitites
[quantities = 0]
[quantities >=0]
Add to Cart
[No]
[Yes]
Add product to cart

3 Detailed Requirements

3.1 Use case Add/Update Product

Use Case "Add/Update Product (Product Manager)"

1. Use case code

UC001

2. Brief Description

This use case describes how a Product Manager adds a new product or updates an existing product in AIMS. The system enforces product-type requirements (Book/Newspaper/CD/DVD), validates the price band (current price must be 30%–150% of the original value), and stores a history of product additions and edits. If an input is invalid, the system notifies the Product Manager.

3. Actors

3.1. Product Manager

4. Preconditions

- The Product Manager is logged in with the appropriate role to access product-maintenance features.
- For Update: the target product already exists in the catalog.

5. Basic Flow of Events (Success)

1. Product Manager requests to Add a new product or Update an existing product.
2. Software displays the product form with a product type selector (Book, Newspaper, CD, DVD).
3. Product Manager enters product data, including title/category (AIMS supports search by product title or category), original value, current price, and type-specific fields as

required.

4. Software validates:
   - Price policy: current price is within 30%–150% of the original value.
   - Type-specific required fields are present (per product type list below).
   - Date formats are valid where applicable (e.g., publication date). If invalid, the system notifies the Product Manager.
5. Software performs the requested action:
   - Add: creates a new product record.
   - Update: applies the submitted changes to the existing record.
6. Software stores a history entry for the addition or edit.
7. Software confirms success and shows the saved product summary.

6. Alternative Flows

Table 2 — Alternative flows of events for UC "Add/Update Product"

No	Location	Condition	Action	Resume location
1	At Step 4	Price violates the 30%–150% band	Software notifies invalid price; submission rejected	Step 2
2	At Step 4	Missing type-specific required info	Software notifies invalid/insufficient info	Step 2
3	At Step 4	Invalid date format (e.g., publication date)	Software notifies invalid format	Step 2
4	At Step 3 (Update)	Product Manager adjusts stock quantity manually	Software requires reason for stock change and records it	Step 4

7. Input data

Table A — Input data for Add/Update Product

A.1 Common fields (all products)

No	Data fields	Description	Mandatory	Valid condition / Notes
1	Title	Title of products	Yes	Title exists
2	Category / Product type	(Book, Newspaper, CD, DVD)	Yes	One of the supported product types
3	Original value	Implicitly inferred from the requirements	Yes	> 0; used for price-band validation (see A.6)
4	Current price	The current price after adjustments from the original price	Yes	Must be within 30%–150% of original value
5	Stock quantity (if updating quantity)		Conditional (Update)	Manual adjustments must record a reason

A.2 Book-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
1	Author(s)	Book authors	Yes	Non-empty	“Robert C. Martin”
2	Cover type	Paperback or hardcover	Yes	One of {Paperback, Hardcover}	Hardcover

3	Publisher	Publishing house	Yes	Non-empty	Pearson
4	Publication date	Date of publication	Yes	Valid date format (the system notifies if date format is incorrect)	30/12/2018

A.3 Newspaper-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
1	Editor-in-chief	Lead editor	Yes	Non-empty	Nguyen Van A
2	Publisher	Publishing organization	Yes	Non-empty	Vietnam News Agency
3	Publication date	Date of issue	Yes	Valid date format (the system notifies if date format is incorrect)	05/10/2025
4	Issue number	Issue identifier	No	—	342
5	Publication frequency	Daily/Weekly/etc.	No	—	Weekly
6	ISSN	International Standard Serial Number	No	—	0868-1234
7	Language	Publication language	No	—	Vietnamese
8	Sections	e.g., politics, business, sports, culture	No	—	Politics; Business

A.4 CD-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example

1	Artists	Performer(s)	Yes	Non-empty	“Adele”
2	Record label	Label/publisher	Yes	Non-empty	XL Recordings
3	Tracks list	List of tracks	Yes	Each track has title and length	“Track 01 – Hello – 4:55”
4	Genre	Musical genre	Yes	Non-empty	Pop
5	Release date	Date of release	No	—	20/11/2015

A.5 DVD-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
1	Disc type	Blu-ray or HD-DVD	Yes	One of {Blu-ray, HD-DVD}	Blu-ray
2	Director	Film director	Yes	Non-empty	Christopher Nolan
3	Runtime	Total running time	Yes	Positive duration	152 min
4	Studio	Production studio	Yes	Non-empty	Warner Bros.
5	Language	Audio language(s)	Yes	Non-empty	English
6	Subtitles	Subtitle language(s)	Yes	Non-empty	English; Vietnamese
7	Release date	Date of release	No	—	16/07/2010
8	Genre	Film genre	No	—	Action

Validation note: When product data **does not comply with rules** or **date format is incorrect**, the software **notifies** the Product Manager.

8. Output data

Table B — Output data after successful Add/Update

No	Data fields	Description
1	Product title & type	The saved product and its type (Book/Newspaper/CD/DVD)
2	Original value & current price	Confirmed values satisfying the 30%–150% band
3	Type-specific attributes	The saved attributes according to the product type (see A.2–A.5)
4	History entry recorded	Addition/edit stored in history

9. Postconditions

- Add: A new product exists in the catalog; a history entry is recorded.
- Update: The product reflects submitted changes; a history entry is recorded.
- Saved current price respects the 30%–150% policy; required type-specific attributes are present.

3.2 Use case Pay Order by VietQR

Use Case "Pay Order by VietQR"

1. **Use case code**
   UC004
2. **Brief Description**
   This use case describes the interaction between the Customer and the AIMS system when the customer wishes to pay for a placed order using a QR code (VietQR).

3. **Actors**

3.1. Customer

3.2. VietQR

3.3. AIMS System

4. **Preconditions**

1. Customer has successfully placed an order and entered valid delivery information

2. The cart is valid, and stock is sufficient for all selected items.

3. The invoice (product list, VAT, and shipping fees) has been successfully generated

4. The AIMS system is connected to the VietQR sandbox environment.

5. **Basic Flow of Events**

6. The software displays the payment options and invoice to the Customer, including product list, quantity, total price, VAT, and delivery fee.
7. The Customer chooses a payment method.
8. The software checks the selected method. If [scan QR method = true], the system continues this use case.
9. The software sends a request to VietQR to generate a payment QR code, containing order information (order ID, total amount, merchant code, transaction content).
10. VietQR generates the QR code and returns the encoded image and transaction reference to AIMS software.
11. The software displays the generated QR code on the screen for the Customer to scan.
12. The Customer scans the QR code using a banking app and confirms the payment transaction.
13. The software verifies the payment transaction status through the VietQR callback or query API.
14. The software evaluates the transaction result
15. The software updates the order status to *Pending Processing*, records the transaction, and sends the payment confirmation email to the Customer.
16. The software displays a message confirming successful payment and summarizes order and transaction information.

6. Alternative flows

Table N-Alternative flows of events for UC Pay order by QR

No	Location	Condition	Action	Resume location

1	Step 4	VietQR API fails to generate QR code.	▪ Display “Unable to generate QR code.” ▪ Allow retry or switch to another payment method.	Step 2
2	Step 7	Customer cancels or closes the payment app before confirming.	▪ Display “Payment not completed.” ▪ Offer retry option.	Step 2
3	Step 8	Payment callback not received (timeout or network error).	▪ Retry verification after a fixed interval. ▪ If still not confirmed, mark order as “Awaiting Payment.”	Use case ends
4	Step 9	Payment verification fails (invalid signature or mismatch).	▪ Reject transaction and notify customer. ▪ Mark order as “Payment Failed.”	Use case ends
5	Step 10	Inventory update or database error	▪ Log the error and set order as “Pending Manual Review.”	Use case ends

7. Input data

Table A-Input data of UC004

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Order ID	Unique identifier of the customer's order	Yes	Alphanumeric	ORD-2025-1008-(
2.	Total amount	Total payable (subtotal + VAT + shipping)	Yes	≥ 0	215,000 VND
3.	Merchant code	Store identifier for VietQR	Yes	String	AIMS-HUST-001
4.	Transaction content	Description for payment record	Yes	$\leq$ 255 chars	Payment for Orde ORD-2025-10 012
5.	Customer info	Customer name, phone, email, address	Yes	Valid format	Nguyen Van A, 0901234567, Hanoi
6.	QR content	Encoded payment string returned from VietQR	Yes	Must follow VietQR format	https://api.vietqr.v abc123xyz

Output data

Table B-Output data of UC004

No	Data fields	Description	Display format	Example
1.	QR code image	Image for customer to scan	PNG/Base64	
2.	Payment confirmation message	Text showing payment success	Text	"Payment complete successfully via VietQR."

3.	Transaction details		Text / Table	TXN-998877 – 215, VND – 2025-10-08 18:36
4.	Order status	Update order state	Text	“Pending Processing
5.	Confirmation email	Invoice and transaction summary sent to customer	Email (HTML/PDF)	

9. Postconditions

- Payment transaction is successfully confirmed through VietQR.
- Order status is updated to Pending Processing.
- Product inventory is decreased according to the order quantity.
- The customer receives an email with the invoice and payment confirmation.
- AIMS stores the transaction record for reconciliation and auditing purposes.

3.3. Use case Pay Order by Credit Card

Use Case “Pay Order by Credit Card”

1. **Use case code**
   UC005
2. **Brief Description**
   This use case describes how the Customer completes an online payment through PayPal (credit card) in the AIMS system.
3. **Actors**

3.1. Customer

3.2. VietQR

3.3. AIMS System

4. **Preconditions**

1. The customer has completed checkout and is viewing a valid invoice.
2. Stock availability and total payable amount are confirmed.
3. The AIMS system is configured with valid PayPal Client ID/Secret and connected to the PayPal Sandbox API.
4. Network connectivity between AIMS and PayPal is stable.

5. Basic Flow of Events

1. The software displays payment options and invoice.
2. The Customer selects Credit Card (PayPal) as the payment method.
3. The software validates order data and reserves stock.
4. The software obtains a PayPal OAuth token and creates a PayPal order
5. PayPal returns an approval URL.
6. The software redirects the Customer to PayPal.
7. The Customer logs in or enters card details and approves payment.
8. PayPal redirects the Customer back to AIMS with transaction info.
9. The software verifies and captures the PayPal order.
10. If capture successful, AIMS stores transaction data, updates stock and order status, and sends a confirmation email.
11. If capture failed, AIMS cancels the draft order and releases reserved stock.
12. The software displays payment result to the customer.

6. Alternative flows

Table N-Alternative flows of events for UC Pay order by QR

No	Location	Condition	Action	Resume location
1	Step 7	Customer cancels payment or closes PayPal before approval.	▪ Display “Unable to generate QR code.” ▪ Allow retry or switch to another payment method.	Step 2
2	Step 4	PayPal API or token request fails.	▪ Display “Payment not completed.” ▪ Offer retry option.	Step 2

3	Step 9	Capture not completed or declined.	▪ Display “Payment unsuccessful.” ▪ Roll back draft order and release stock.	Step 2
4	Step 10	Error while updating order or stock.	▪ Log error, mark order “Pending Manual Review.”	Use case ends

7. Input data

Table A-Input data of UC005

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Order ID	Unique order identifier	Yes	Alphanumeric	ORD-2025-1008-0
2.	Total amount	Sum of items + VAT + shipping	Yes	≥ 0	215,000 VND
3.	Customer info	Name, phone, email, address	Yes	Valid format	Nguyen Van A, 0901234567, Hanoi
4.	PayPal OAuth Token	Authentication token	Yes	Valid JWT format	A21AAKQ...token
5.	Approval URL	Redirect link returned by PayPal	Yes	HTTPS URL	https://www.sandbaypal.com/...
6.	Capture ID	Identifier of successful capture	Yes	Alphanumeric	CAP-9AB23952R34567

8. Output data

Table B-Output data of UC005

No	Data fields	Description	Display format	Example
1.	Approval link	Redirect URL for PayPal checkout	Hyperlink	
2.	Transaction status	Text showing payment success	Text	“Payment complete successfully via Paypal.”
3.	Transaction info	Capture ID, amount, payer, time	Table	CAP-9AB23952RC 4567 – 325,000 VN 2025-10-08 18:42
4.	Order status	Update order state	Text	“Pending Processing
5.	Confirmation email	Invoice and transaction summary sent to customer	Email (HTML/PDF)	

9. Postconditions

- The PayPal transaction is successfully captured and recorded.
- Order status is updated to Pending Processing.
- Inventory quantities are adjusted.
- Confirmation email is sent to the customer.
- In case of failure or cancellation, the system releases stock and keeps an audit log.

3.4. Use case Place Order

Use Case “Place Order”

1. Use case code

UC002

2. Brief Description

This use case describes how a Customer finalizes the purchase of items in their cart. It covers the entire process from validating the cart, providing shipping information, processing payment, and receiving final confirmation for the order.

3. Actors

3.1.Customer

4. Preconditions

- The Customer's cart contains at least one item.

5. Basic Flow of Events (Success)

1. The Customer requests to place an order from the cart screen.
2. The System checks if the stock quantity is sufficient for each product in the cart.
3. The System displays the delivery information form.
4. The Customer enters and submits their delivery information.
5. The System validates the delivery information.
6. The System calculates the shipping fee based on the total weight and delivery location, applying any applicable promotions
7. The System displays the final invoice screen, showing product prices, VAT, shipping fee, and the total amount.
8. The System calls the "Pay Order" use case.
9. After a successful return from the "Pay Order" use case, the System displays a confirmation screen with order details and transaction information. The screen presents two options: "Confirm Order" and "Cancel Order".
10. The Customer reviews the information and selects "Confirm Order".
11. The System saves the new order with a "pending" status and records the payment transaction.

12. The System sends a detailed order confirmation to the Customer's email.
13. The System displays a final "Order Successful" screen.

6. Alternative Flows

Table 1 — Alternative flows of events for UC “Place Order”

No	Location	Condition	Action	Resume location
1	At Step 2	The requested quantity for an item exceeds the available stock.	The System notifies the Customer which item is out of stock and displays the available quantity. The System prompts the Customer to update the cart.	Use case ends.
2	At Step 5	The delivery information is invalid (e.g., blank required fields, phone number format is incorrect).	The System displays an error message next to the invalid fields and prompts the Customer to correct them.	Step 4
3	At Step 8	The "Pay Order" use case returns unsuccessfully (payment failed).	The System displays a "Payment Failed" notification and allows the Customer to try again.	Step 7
4	At Step 10	The Customer selects "Cancel Order" on the final confirmation screen.	The System immediately processes a full refund using the original payment method, sends a cancellation confirmation email, and displays a	Use case ends.

			cancellation success message.	

7. Input data

Table 2 — Input data for Delivery Information

No	Data fields	Description	Mandatory	Valid condition	Example
1	Receiver Name	Full name of the recipient	Yes	Not blank	Nguyen Van A
2	Phone Number	Contact phone number	Yes	10 digits, starts with '0', may contain '.', '-', ' '	0987-654 -321
3	Province/City	Delivery location	Yes	Selected from a predefined list	Hanoi
4	Address	Street address details	Yes	Not blank	1 Dai Co Viet, Hai Ba Trung District
5	Shipping Instructions	Special notes for delivery	No	-	Please call before delivery.

8. Output data

Table 3 — Output data for Invoice Screen Data

No	Data fields	Description	Display format	Example
1	Product Title	Title of a media product	-	AIMS DVD
2	Price	Price per unit (including VAT)	Comma separator, VND	110,000 VND
3	Quantity	Quantity of the item	Integer	2
4	Amount	Total price for the item	Comma separator, VND	220,000 VND
5	Subtotal	Total amount for all products	Comma separator, VND	450,000 VND
6	Shipping Fee	Calculated delivery fee	Comma separator, VND	22,000 VND
7	Total	Final amount to be paid	Comma separator, VND	472,000 VND

Table 4: Successful Order Screen Data

No	Data fields	Description
1	Customer Name	Name of the person who ordered
2	Phone Number	Contact phone number
3	Shipping Address	Full delivery address
4	Total Amount	Total amount paid

9. Postconditions

- On successful completion: A new order is created in the system with a "pending" status. A payment transaction is recorded. A confirmation email is sent to the Customer.
- On cancellation or failure: No order is created in the system. If payment was processed and then cancelled, a refund transaction is initiated.

3.5. Use case View Product Details

Use Case “View Product Details”

1. Use case code

UC003

2. Brief Description

This use case describes the interaction between Customer and AIMS when the customer wants to view the detailed information of a selected product (Book / CD / DVD / Newspaper) and may choose a quantity and add the product to the cart.

3. Actors

Customer

4. Preconditions

The customer has navigated to the product list/search results and selected a product. No login is required.

5. Basic Flow of Events (Narrative)

1. The customer selects a product from the list/search results to view its details.
2. The system checks whether the selected product exists and is available to be shown.
3. If the product exists, the system determines the product type.
4. Based on the identified type, the system retrieves the type-specific details

together with the common information of the product.

5. The system displays the detailed information of the selected product.
6. The customer reviews the details and selects a corresponding quantity.
7. If the selected quantity is zero, the use case ends.
8. If the selected quantity is greater than zero, the customer chooses Add to Cart and confirms the action.
9. When confirmed, the system adds the product to the cart with the selected quantity and the use case ends.
10. If the customer cancels the add action, the use case ends with no change to the cart.
11. If, at Step 2, the product does not exist, the system shows "Product not Found" and the use case ends.

6. Alternative Flows

Table 1 — Alternative flows of events for UC "View Detail Product"

No	Location	Condition	Action	Resume location
1	At Step 2	productId not found	Software shows "Product not found"	End
2	At Step 6	Selected quantity equals 0	No item is added to cart	End
3	At Step 8	Customer cancels the add-to-cart action	No item is added to cart	End

7. Input data

None

8. Output data

Table A – Common information of selectied product

No	Data fields	Description	Display Format	Example
1	Title	Name of product	String	Introduction to Artificial Intelligence
2	Price	Price of product	Non-negative integer	150000
3	Category	Category of product	String	book
4	Avail	Number of products available in store	Non-negative integer	12
5	Image	Image of product	Image (.png / .jpg)	book.png

Table B – Information of Book

No	Data fields	Description	Display Format	Example
1	Author	Author of the book	String	Andrew Ng
2	Publisher	Publisher of the book	String	MIT Press
3	Cover Type	Type of the cover	String	HardCover
4	Number of Page	Number of pages	Positive integer	420
5	Publication Date	Date when the book is published	Date	2025-02-27
6	Language	Language of book content	String	English

7	Genre	Genre type of book	String	Computer Science

Table C – Information of DVDs

No	Data fields	Description	Display Format	Example
1	Disc Type	Type of DVD	String	Blu-Ray
2	Director	Director of the movie	String	Christopher Nolan
3	Studio	Studio producing the movie	String	Warner Bross
4	Language	Primary spoken language in the movie	String	English
5	Subtitle	Available subtitle languages	String	Vietnamese, Japanese
6	Release Date	Date when the DVD was released	Date	2024-10-20
7	Genre	Genre of the movie	String	Science Fiction

Table D – Information of CDs

No	Data fields	Description	Display Format	Example

1	Artist	Name of artist or band	String	Taylor Swift”
2	Record Label	Music record label	String	Republic Record
3	Genre	Music genre	String	Pop
4	Release Date	Date when the album was released	Date	2022-11-18
5	Track list	List of songs in the CD (each includes title and duration)	String	1. Lavender Haze – 3:22; 2. Anti-Hero – 3:20; 3. Karma – 3:25

Table E – Information of Newspapers

No	Data fields	Description	Display Format	Example
1	Editor-in-chief	Chief editor responsible for publication	String	Pham Thanh Tung
2	Publisher	Publisher or press agency	String	Vietnam News Agency
3	Publication Date	Date of publication	Date	2025-03-05
4	Issue Number	Newspaper issue number	Non-negative integer	315
5	Publication Frequency	Frequency of release	String	Daily

6	ISSN	International Standard Serial Number	String	2616-0455
7	Language	Publication language	String	Vietnamese
8	Sections	Newspaper sections	String	Politics; Business; Technology; Culture; Sports

9. Postconditions

- The system has displayed the detailed information of the selected product.
- If the customer confirmed adding with a positive quantity, the product has been added to the cart with that quantity; otherwise, the cart remained unchanged.

4 Supplementary specification

4.1 Functionality

- **Authentication:** Distinguish public access (Customer, no login) from authenticated access (staff accounts).

- **Authorization (RBAC):** Role-based access control for Administrator/Product Manager features; least-privilege by default.

- **Validation:** Consistent data/business-rule checks at system boundaries (e.g., required fields, price bands, dates).

- **Error Handling & User Feedback:** Uniform error taxonomy, safe messages, and recoverable flows (retry, correct-and-resubmit).

- **Auditing & Logging:** Structured logs for actions/events and an audit trail for catalog changes and order lifecycle.

- **Transactions & Consistency:** Atomic operations for multi-step updates (e.g., order/payment/state changes); rollback on failure.

- **Idempotency & Concurrency:** Idempotent endpoints for retried requests (e.g., payments); protect against concurrent edits.

- **Security (Data Protection):** Secure handling of credentials/tokens, transport encryption, safe storage of sensitive fields.

- **Notifications:** Cross-cutting email (and future channels) for key events (order placed/cancelled, admin actions).

4.2 Usability

- **Task continuity during ordering**: Users can **return to any step** or exit the flow while ordering; the system preserves state as specified.

- **Clear validation & feedback**: When inputs violate rules or formats, the system **notifies** the user performing the action (e.g., PM during product maintenance; customer during cart/order).

- **Readable monetary presentation**: Amounts shown on invoices and carts use **thousand separators**, positive integer display, and **right alignment**; totals show **excl./incl. VAT** and shipping as specified.

- **Discoverability**: Product discovery supports browsing/search (title/category) and price filters; details are shown by type (supports learnability without training). AIMS-ProblemStatement-ver3.0

- **Documentation tone**: Per slides, keep requirements **UI-agnostic** in this SRS; UI specifics appear only when necessary to understand behavior.

4.3 Reliability

- **Availability**: The system operates for ≥ **300 hours without failure** (High availability).

- **Fault recovery**: On failure, **maximum recovery time ≤ 1 hour**.

- **Load tolerance**: Handles **1,000 concurrent users** without breaching performance targets.

- **Data integrity**: Operations that change state (orders, payments, product edits) are **atomic**; partial failures result in rollback to a consistent state.

4.4 Performance

- **End-to-end latency**: Typical responses **≈ 2 seconds**; at peak load, **≤ 5 seconds**.

- **Throughput at scale**: With **1,000 CCUs**, the system continues to meet the above response targets.

- **Ordering flow responsiveness**:

  - **Stock checks** on "place order" requests complete before advancing the flow.

  - **Shipping fee calculation/recalculation** occurs **whenever address changes**; invoices update accordingly.

- **Pagination for staff workflows**: **Pending orders: 30 per page**, ensuring predictable render time under normal load.

4.5 Supportability

- **Extensible payments**: Integrations with **VietQR** (QR) and **PayPal Sandbox** (credit card) are modular to allow **future methods** without core changes.

- **Extensible notifications**: Email is required now; **SMS/push** may be added later without altering business logic.

- **Role and user lifecycle**: Admin features (create/update/delete users, reset passwords, block/unblock, assign roles) support routine maintenance and onboarding.

- **Specification hygiene**: Non-functional requirements (this section) capture cross-cutting behavior **not tied to a single use case**, per slides' Supplementary Spec guidance.

4.6 Other requirements

- **Security & access control**:

  - **No login** for customers; **login required** for Product Managers and Administrators.

  - **Role-based authorization**: roles can be **multiple per user**; restrict staff features accordingly.

- **Auditability**: The system **stores a history** of product additions/edits/deletions; staff actions that change order state are logged.
  AIMS-ProblemStatement-ver3.0

- **Regulatory/formatting**: **Shipping fees not subject to tax**; invoices show **excl./incl. VAT** totals clearly.

- **Error handling**: For insufficient stock or invalid inputs, the system **notifies** users and supports correction/resubmission (customer can retry order; PM can correct product data).